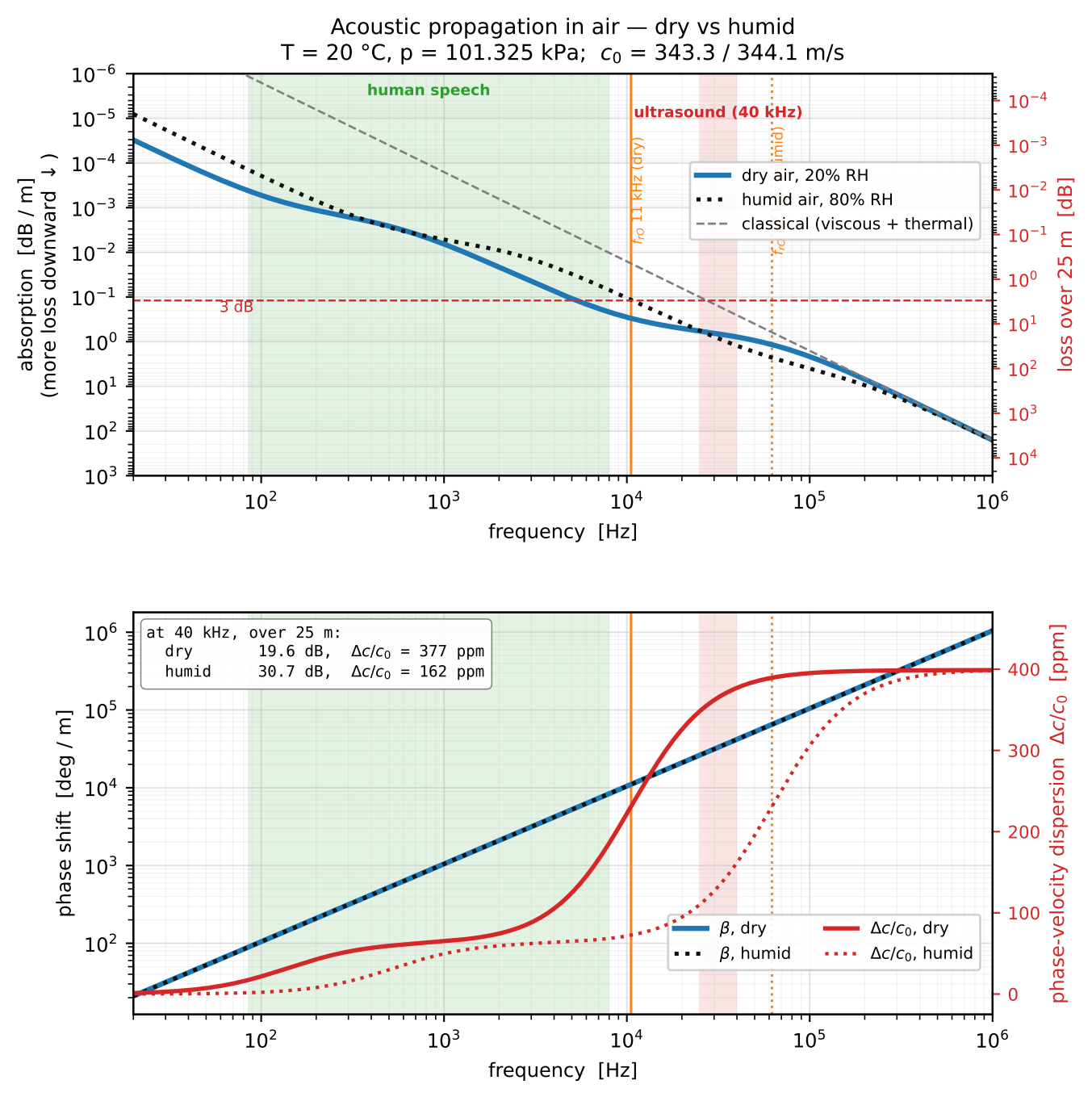
Acoustic propagation in air — dry vs humid
T = 20 °C, p = 101.325 kPa; c_0 = 343.3 / 344.1 m/s
human speech
ultrasound (40 kHz)
f_{rO} 11 kHz (dry)
f_{rO} (humid)
3 dB
dry air, 20% RH
humid air, 80% RH
classical (viscous + thermal)
absorption [dB / m]
(more loss downward ↓)
loss over 25 m [dB]
frequency [Hz]
at 40 kHz, over 25 m:
dry 19.6 dB, Δc/c0 = 377 ppm
humid 30.7 dB, Δc/c0 = 162 ppm
phase shift [deg / m]
phase-velocity dispersion Δc/c0 [ppm]
β, dry
β, humid
Δc/c0, dry
Δc/c0, humid
frequency [Hz]